AI Prompting in 2026: A Crash Course

Foundations (Everyone), Chapter 3/10 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/ai-prompting-2026

EN: The model is stateless -- almost every 'advanced technique' is one move in two disguises: get the right context IN, keep the wrong context OUT. 13 concepts across knowing, talking, beyond text, and working safely.
Roman Urdu: Model stateless hai -- almost har 'advanced technique' ek move hai do disguises mein: sahi context andar lao, ghalat context bahar rakho. 13 concepts: knowing, talking, beyond text, safely working.

1. How AI Knows Things (Concepts 1-3)

- Mental model: AI = a smart-but-new colleague who knows nothing about you yet. Pretrained knowledge: strong (popular topics), sparse (niche/regional), absent (private data, post-cutoff).
- **3 retrieval modes:** pretrained, web search (stale-rescue, doesn't check source currency), deep research (dozens of sources, multi-section report).

2. Talking to AI Well (Concepts 4-7)

- **Context is the whole game** -- the window isn't empty (system prompt pre-loaded). Context rot: switch topics, start a new conversation. Projects/Notebooks front-load context once.
- **Reasoning ('think hard'):** invoke built-in reasoning modes explicitly for hard tasks; skip for quick lookups.
- **Sycophancy:** models are agreement-biased (~10x 'yes' vs 'no'). Avoid bait verbs (find/defend/confirm); use neutral ones (evaluate/compare/critique). Force a number score.
- **Brainstorm-Iterate Loop** (highest-leverage habit): give context -> ask for 3-5 options -> give explicit feedback -> new options -> iterate -> then detail.

3. Beyond Text (Concepts 8-10)

- Multimodal: image input strong on scenes/whiteboards/handwriting, weak on fine detail/counting. Power-user recipe: SVG -> PNG -> redraw -> iterate.
- Small apps: Goal/Input/Output recipe; Artifact = persistent editable object.
- Data analysis: the model writes/runs code -- it CHOOSES to, doesn't guarantee it. Explicitly demand verifiable, specific output.

4. Working Safely + Choosing Tools (Concepts 11-13)

- Desktop apps (Cowork/OpenWork): 2 hard facts -- deleted files skip the recycle bin, no version history. Safe workflow: task -> plan (not action) -> review -> approve.
- AI is jagged -- try 2-3 models on the same prompt; check leaderboards monthly.
- **Models checking models:** use a different model family to grade output when there's no ground truth. Caveat: a score is a progress signal, not a truth signal -- high-stakes work still needs a human expert.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Stateless model	The model carries nothing between conversations by itself.	Model khud conversations ke beech kuch carry nahi karta.
Context rot	Context quality degrades as topics pile up in one conversation.	Topics jama hone se conversation ka context quality girta hai.
Sycophancy	The model's bias toward agreeing with you.	Model ka aap se agree karne ki taraf bias.
Brainstorm-Iterate Loop	Context -> options -> feedback -> new options -> detail.	Context -> options -> feedback -> new options -> detail.
Jagged frontier (applied)	Try 2-3 models on the same prompt -- capability varies unevenly.	Same prompt 2-3 models par try karo -- capability unevenly differ karti hai.
Cross-model grading	Using a different model family to score output when there's no ground truth.	Ground truth na ho to doosre model family se output grade karwana.

Quick Self-Check

Question	Answer
Why does context rot happen, and what's the fix?	Switching topics inside one long conversation degrades quality -- start a new conversation instead.
What are bait verbs, and why avoid them?	Verbs like find/defend/confirm/prove push the model toward agreeing; use neutral ones like evaluate/compare/critique.
What's the 6-step Brainstorm-Iterate Loop?	Give context, ask for 3-5 options, give explicit feedback, get new options, iterate, then add detail.
Why must desktop AI apps (Cowork/OpenWork) show a plan before acting?	Deleted files don't go to a recycle bin and there's no version history -- mistakes aren't undoable.
Is a cross-model grading score a truth signal?	No -- it's a progress signal; high-stakes (legal/medical/financial) work still needs a human expert.

Where This Leads / Sources

Leads to: 4 practice projects (Snake Battle, Whack-a-Mole, A Page That Is You, AI Mini Textbook capstone). Full 12-prompt recap + 30-Q interactive quiz on the live book page.

Source: docs/ai-prompting-2026/ (this repo), content originally via Zia Tutor AI connector.